THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** magnet, add water and filter, evaporate filtrate	Each step removes one component.
2	**(b)** it has no cells and no growth or reproduction	Many characteristics are absent.
3	**(d)** most oxygen was used	Burning consumed oxygen.
4	**(a)** 8 light-minutes	Approx.
5	**(a)** 30 g	Same as initial.
6	**(b)** bacterium	It has one cell.
7	**(a)** adding compost	Compost adds humus.
8	**(d)** 60 kg	Mass does not change.
9	**(b)** it makes tiny particles clump and settle	Coagulation.
10	**(d)** a living thing can stay inactive until conditions suit it	Life processes were only slowed down.
11	**(b)** more public transport	Fewer vehicles.
12	**(b)** Earth's axis is tilted as it goes around the Sun	Tilt.
13	**(a)** sand	Insoluble.
14	**(b)** the crystal grows by outside deposition, not by cell division	Non-living.
15	**(a)** oxygen forms about one-fifth of air	Oxygen is used up.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Some salt stuck to the filter or vessel (1); Handling losses (1).	1+1
17	It does not grow or reproduce (1); It does not respire or respond (1).	1+1
18	Oxygen about 21 per cent is used (1); Water enters to fill that space (1).	1+1
19	228/150 (1); about 1.5 cm (1).	1+1
20	Dissolve the sugar in water and strain out the grains (1); Evaporate the water to recover sugar (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Magnet for iron (1); Water: wooden chips float, skim (1); Filter sand and chalk; evaporate salt (1).	1+1+1
22	No chlorophyll (1); Do not make their own food (1); Absorb food from dead matter (1).	1+1+1
23	21 per cent of 50 (1); 0.21 x 50 (1); 10.5 m^3 (1).	1+1+1
24	5000/10 (1); 500 days (1); Year/day (1).	1+1+1
	OR: Immiscible (1); Kerosene lighter (1); Floats on top (1).	1+1+1
25	Respiration is very slow (1); It germinates later (1); So it is alive but dormant (1).	1+1+1
	OR: Half the sheets (1); 100 sheets (1); 100 saved (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Keep both still (1); Mud settles, salt water stays clear (1); Filter both (1); Mud on paper, salt in filtrate (1); Evaporate filtrate for salt (1).	1+1+1+1+1
	OR: Set A: dry seeds (1); Set B: seeds in water without air (boiled water with oil) (1); Set C: seeds in moist cotton at room temperature (1); Set D: moist seeds in the fridge (1); Only C germinates well (1).	1+1+1+1+1
27	List water use (1); List electricity use (1); Record fuel (1); Identify wastage (1); Suggest five actions (1).	1+1+1+1+1

	OR: Choose a scale for sizes (1); Place the Sun at the centre (1); Planets in order (1); Label features (1); Show orbits (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** waxing	Increasing.
2	**(b)** 18 g	Half.
3	**(b)** water is needed for germination	Controlled comparison.
4	**(b)** loosening and mixing it	Aeration.
5	**(b)** one sixth	Weaker gravity.
6	**(c)** filtration cannot remove all germs	Disinfection is needed.
7	**(a)** change in body during growth	Life cycle.
8	**(b)** planting a tree for every tree cut	Balance.
9	**(a)** stars appear to move across the night sky	Apparent motion.
10	**(c)** fine clay	Smaller particles settle slowest.
11	**(a)** it swells and sprouts in water	Growth suggests life.
12	**(a)** nitrogen > oxygen > carbon dioxide	78, 21, 0.04.
13	**(a)** 30 March	About 29.5 days.
14	**(a)** 7 g	25 - 18.
15	**(a)** type of seeds and temperature	Same seeds and temperature.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Cells have died (1); No respiration or growth (1).	1+1
17	Take in carbon dioxide (1); Give out oxygen (1).	1+1
18	They orbit the Sun (1); Closer than the stars (1).	1+1
19	Strain through cloth (1); Warm gently (1).	1+1
20	River, clouds (1); Car, wind (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Gravity pulls planets (1); Pull bends their path (1); Without it they would fly off in straight lines (1).	1+1+1
22	Holds solid (1); Whey passes (1); Straining (1).	1+1+1
23	Plants give oxygen (1); Animals give CO2 (1); Bees pollinate flowers (1).	1+1+1
24	Steep slopes (1); Fast runoff (1); Afforestation (1).	1+1+1
	OR: Moon is a quarter of the way around (1); Half of the lit side visible (1); Looks like a half disc (1).	1+1+1
25	36 g in 100 g (1); 90/36 x 100 (1); 250 g water (1).	1+1+1
	OR: Two flasks, one with germinating seeds, one with boiled seeds (1); Insert thermometers (1); After hours the former is warmer (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Sea water collects in pans (1); Evaporation in stages (1); Brine thickens (1); Crystals formed (1); Washed and dried (1).	1+1+1+1+1
	OR: Roots to water (1); Shoots to light (1); Camouflage (1); Fleeing predators (1); Hibernation or migration (1).	1+1+1+1+1
27	Reduce plastic (1); Walk or cycle (1); Save water (1); Recycle (1); Plant trees (1).	1+1+1+1+1
	OR: Moon's phase depends on position (1); Not on place (1); Rise times differ with longitude (1); Earth rotates (1); Same shape seen (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** erode	No cover.
2	**(b)** 16	1440/90.
3	**(a)** sedimentation, filtration, disinfection	Standard.
4	**(b)** accumulated wastes are toxic	Removal of harmful substances.
5	**(b)** no fuel cost and no smoke	Clean.
6	**(c)** halve	12 hours.
7	**(d)** separating funnel	Layers form.
8	**(a)** movement	Machines and rivers move.
9	**(b)** less soil washed away	Conservation.
10	**(b)** sunset	Opposite the Sun.
11	**(d)** 20 g	Half of 40.
12	**(b)** increase in number of cells	Cell division.
13	**(a)** nutrient loss in soil	Monoculture.
14	**(d)** 467	42000/90.
15	**(d)** to avoid disturbing the sediment	Keeps the sediment.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	They break down wastes (1); Mix and aerate soil (1).	1+1
17	Earth's rotation speed is highest (1); Gives extra boost (1).	1+1
18	Sieve holes are large (1); Filter paper holes are tiny (1).	1+1
19	One feature may be seen in non-living things (1); Combination makes the test reliable (1).	1+1
20	Particular nutrients are used up (1); Crop rotation helps (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	60/15 (1); 4 L (1); Saves fuel and avoids pollution (1).	1+1+1
22	365/27 (1); About 13.5 (1); Approximately 13 complete orbits (1).	1+1+1
23	Separates whey (1); Solid curd stays (1); Liquid passes (1).	1+1+1
24	Plants respond to gravity (1); Shoots grow upward (1); Roots grow downward (1).	1+1+1
	OR: Burning produces toxic gases (1); Burying fills land (1); Both harmful; reduce use (1).	1+1+1
25	Long orbits (1); Take decades or centuries (1); Appear near the Sun (1).	1+1+1
	OR: Add water (1); Oil floats (1); Filter sand (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Let sand settle, decant (1); Filter remaining (1); Dry the sand (1); Evaporate part of the filtrate for salt (1); Condense vapour for water (1).	1+1+1+1+1
	OR: Dry seed: inactive but germinates (1); Crystal: grows from outside (1); River: moves but does not grow or respire (1); Need many criteria (1); Cells decide (1).	1+1+1+1+1
27	Soil erosion (1); Landslides (1); Less rain (1); Loss of wildlife (1); Water scarcity (1).	1+1+1+1+1
	OR: New moon starts the cycle (1); First quarter day 8 (1); Full moon day 15 (1); Last quarter day 22 (1); Cycle about 29.5 days (1).	1+1+1+1+1

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